A unified model for etched-track formation in CR-39 detectors using amorphous track structure theory

Jeppe Brage Christensen
jeppe.christensen@psi.ch

Department of Radiation Safety and Security, Paul Scherrer Institute, Forschungsstrasse 111, 5232 Villigen PSI, Switzerland

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Abstract

Most existing track formation models for track-etch detectors relate the local etch-rate v_t to the particle's linear energy transfer (LET) along its trajectory. While this approach reproduces experimental data well, it has fundamental limitations: LET is an average measure of energy loss and does not capture the local energy deposition around the particle trajectory, i.e., the radial dose distribution. As a result, for conventional track formation models, the track etch-rate v_t is derived from experiments for *each* particle type and energy, which is both very demanding and limits predictive capabilities.

To generalise track formation modelling and overcome previous limitations, this work models the local etch-rate as a function of the local absorbed dose d, i.e. v_d, rather than as a function of LET or residual range along the track, v_t. The track contours are calculated in a three iterations: (i) The local dose d around the ion trajectory is modelled through a combination of amorphous track structure theory, which accounts for the radial dose deposition by secondary electrons, and stopping power tables. (ii) Next, a single generalised etch-rate function v_d is identified to model how local polymer damage translates into etching speed for different ion species and energies. Along with the bulk etch-rate v_{bulk} for unirradiated material, this enables the calculation local etch-rates v in the entire detector. (iii) Finally, the track contour is calculated by interpreting the etching front as a wave front propagating through a medium with spatially varying speed, where the local speed is given by the etch-rate v at each point. Iso-contours of equal arrival time correspond to the track contour after a given etching time.

Validation against experimental data for various ion species and energies shows agreement within the uncertainties between predicted and observed track geometries. Unlike LET-based models, this local dose-based model provides predictive capabilities for ions and energies outside the calibration domain. The framework, `tracketch`, is available as an open-source package.

1 Introduction

Track-etch detectors have been used for charged-particle detection and dosimetry since the 1960s (Fleischer et al., 1975). Amongst the possibilities, CR-39, a poly-allyl diglycol carbonate (PADC) polymer, remains one of the most widely used solid-state nuclear track detectors. When a charged particle traverses the detector, it deposits energy along its trajectory, producing a trail of damaged polymer chains known as a latent track. These latent tracks can be revealed by subjecting the detector to a chemical etching treatment. During etching, the damaged regions of the polymer are removed at a higher rate than the undamaged bulk material, leading to the formation of conical etch pits. The morphology of these pits, e.g. their shape and size, contains information about the incident particles (Bolzonella et al., 2022).

The applications of CR-39 detectors span a wide range of fields, including particle identification relevant to space research (Kodaira et al., 2013), measurements of energy spectra (E. R. Benton et al., 2010), out-of-field measurements in particle therapy (Bolzonella et al., 2025), personal neutron dosimetry at accelerator facilities (Schmidt et al., 2025), and radiation protection at nuclear power plants (Mayer et al., 2023). In neutron dosimetry, chemically etched CR-39 detectors are used for their high sensitivity to charged particles produced in neutron-induced reactions. Neutron interactions with a suitable converter material or with the detector itself can generate recoil protons or other ionising particles that leave latent tracks in the CR-39 detector. Depending on the etching conditions, these recoil protons may produce tracks that are visually difficult to distinguish from background noise, as they exhibit similar morphological properties (Bolzonella et al., 2022). To improve background discrimination in such applications, models capable of predicting track morphology for a wide range of particle types and energies are required.

The underlying physical processes leading to track formation in CR-39 have been studied for decades to develop such track formation models. As ions pass through CR-39 detectors, they lose energy primarily via ionization and excitation of the polymer molecules. This energy deposition can be described in terms of secondary electrons (δ-electrons) liberated along the ion's path (Katz, 1991). The resulting radial dose distribution (RDD) around the ion track plays a central role in track formation, as it determines the spatial extent of polymer damage, and thus eventually the local etch-rate. To describe the RDD, amorphous track structure models have been developed to characterize the local dose distribution surrounding ion tracks in various materials (Cucinotta et al., 1998; Kraft et al., 1992).

As early as the 1980s, Robert Katz proposed that the response of CR-39 could be understood in terms of interactions between δ-electrons and polymer clusters acting as sensitive targets (Katz, 1984). Using relatively new dose response models developed for dosimetry and radiobiology, Katz modeled the response of CR-39 detectors as a 1-hit detector, where polymer clusters form the sensitive target. He subsequently proposed a track structure based model to describe the detector response. Although his framework provided a physical basis for track formation, the lack of experimental data at the time limited further development, and the focus remained on detector response.

Extending the early work by Katz, Sermund et al. (1984) used the cylindrical symmetry of ion tracks to introduce the radial etch-rate $v(r)$, i.e. the etch-rate as a function of radial distance from the track core. They demonstrated that the radial etch-rate exhibits a structure similar in shape to the RDDs proposed by Katz. However, no explicit track formation model based on $v(r)$ appears to have been developed.

Other approaches rely on the so-called *two etch-rates*, where the focus shifted from track structure theory to models based on unrestricted LET or restricted LET. These track formation models distinguish between the etch-rate along the particle track v_t and the isotropic bulk etch-rate v_{bulk} (Fromm et al., 1991). The variation of v_t with the restricted or unrestricted LET has been investigated since the 1960s (E. V. Benton & Nix, 1969; Fleischer et al., 1975; Fromm et al., 1991). Several track formation models have been developed based on empirical relationships between v_t and v_{bulk} (Azooz et al., 2012; Dörschel et al., 2002, 2003a; Fromm et al., 1991; Nikezic & Yu, 2004, 2006). While this approach is convenient and often provides good agreement with experimental data, it is fundamentally limited because LET represents an average energy loss per unit path length and does not capture the spatial distribution of energy deposition around the track. Both restricted and unrestricted LET has been used for several decades (E. V. Benton & Nix, 1969; Caresana et al., 2012; Fowler et al., 1979), where restricted LET may be interpreted as a step towards taking the radial dose distribution into account.

Neither unrestricted nor restricted LET, however, can describe the RDD that governs the microscopic polymer damage ultimately responsible for track formation in CR-39. As a consequence, v_t must be experimentally determined for *each particle* species and *each energy*. Hermsdorf (2009) demonstrated how v_t relationships must be established uniquely for protons, deuterons, tritons, and alpha particles to match experimental data with *two etch-rate* models. This means, such models generally lack predictive power for particle types and energies, for which no experimental data is available.

To overcome the limitations of LET-based descriptions of track formation in CR-39 detectors, it is necessary to shift the focus from average energy loss back to the local dose distribution around the ion track, i.e. the RDD, as originally proposed in the 1980s. In this context, more recent studies have examined the distribution of electrons around the ion track and employed Monte Carlo particle transport methods to assess the electron ranges and fluences for which latent tracks may form (Bernal et al., 2015; Fromm et al., 2015; Kusumoto et al., 2020, 2021; Mori et al., 2012; Yamauchi et al., 2021). It has been shown that other metrics than LET, but related to track structure theory, provide a better description of the underlying physical processes of ion interactions with CR-39 (Kusumoto et al., 2018). This is analogous to other types of solid-state detectors, in which other radiation quality metrics than LET are employed to better describe the detector response (Christensen et al., 2024).

Rather than relying on empirical relations of the form $v_t(\mathrm{LET})$, the etch-rate v can be generalized to depend on the local dose d, i.e. $v_d(d)$, enabling a physically motivated description of polymer damage and etching. If the relationship $v_d(d)$ can be established across different ion species, track formation can be predicted for arbitrary particle types and energies by combining the RDD with the corresponding etch-rate model.

The current absence of such a predictive model was already highlighted more than 30 years ago by Enge (1995), who concluded:

> *Since about 30 years etched nuclear tracks in solids are known and there is still no unifying model to understand the formation of the etched tracks of heavy ions.*

This motivates the present work, in which the track structure approach originally proposed by Katz (1984) is revisited and extended to develop, to my knowledge, the first unified track formation model for CR-39 detectors based on amorphous track structure theory.

The aim of this work is to demonstrate that a single etch-rate function, $v_d(d)$, exists and, when combined with amorphous track structure theory, can predict track formation for arbitrary particle types and energies. The model is validated against experimental data from literature and is made available as an open-source toolkit `tracketch`[1].

2 Methodology

2.1 Algorithm overview

The proposed track formation model aims to predict the etch-pit geometry formed in CR-39 detectors after chemical etching, given the type and energy of the incident ion as well as the etching conditions. To develop the model and demonstrate the existence of a generalized etch-rate function $v_d(d)$, this work focuses on ions impinging perpendicularly onto the detector surface. Due to the cylindrical symmetry of this configuration, the model can be reduced to two dimensions: the radial distance r from the track axis and the depth z along the track axis as the ion slows down within the CR-39 detector. While this study is restricted to normal incidence, the model can be extended to arbitrary angles of incidence in future work, *e.g.* simply by rotating the coordinate system after the dose computation.

To predict track formation, the model consists of three main steps, as illustrated in figure 1. Each step is described in detail in the following sections and consists of

(a) **Dose computation**: To model the radial dose distributions $d(r)$ around the ion trajectory, amorphous track structure theory is used in conjunction with stopping power tables to calculate the local dose distribution $d(r, z)$ as the ion slows down within the CR-39 detector (Section 2.2).

(b) **Etch-rate computation**: A single generalised etch-rate model $v_d(d)$, applicable to all investigated ions, is used to convert the local dose distribution into a spatially resolved etch-rate distribution $v(r, z)$ map. Identifying and calibrating an appropriate $v_d(d)$ function constitutes a central component of this work (Section 2.3).

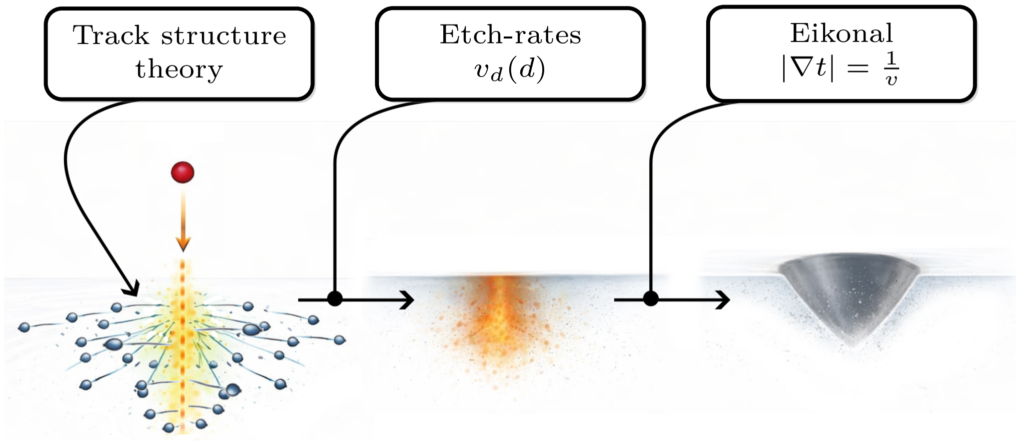
(c) **Track contour calculation**: The etch-rate distribution $v(r, z)$ is interpreted as a speed map for an advancing wave front. The arrival times of this advancing etching front at each point in the detector are computed by solving the eikonal equation (Sethian, 1999). Iso-contours of equal arrival time correspond to the track contour after a given etching time (Section 2.4).

Thus, in this work, $d(r, z)$, $v(r, z)$, and $t(r, z)$ denote scalar fields representing the local dose, local etch-rate, and arrival time of the etching front at each point in the detector, respectively. Each of these steps is detailed in the sections below. The algorithm is implemented in `python` and is made available as an open-source tool (Christensen, 2026).

2.2 Radial dose distribution

Amorphous track structure theory provides a framework to calculate the radial dose distribution (RDD) around an ion track in a given material. Some RDDs commonly used in track structure theory are presented in Kraft et al. (1992) and Cucinotta et al. (1998).

[1] https://github.com/jbrage/tracketch



(a) Local doses $d(r, z)$ (b) Etch-rates $v(r, z)$ (c) Track contours $t(r, z)$

Figure 1: Schematic overview of the track formation model for an ion incident perpendicular to the detector surface. (a) Amorphous track structure theory and stopping power tables are used to calculate the local dose distribution $d(r, z)$ as the ion slows down in the CR-39. (b) Using a calibrated etch-rate model $v_d(d)$, the local etch-rate distribution $v(r, z)$ is obtained. (c) The track contour is calculated by evaluating the arrival times of the etching front at each point in the detector. Iso-contours of equal arrival time correspond to the track contour after a given etching time.

To calculate the RDD for a given ion type and kinetic energy, the open-source library `libamtrack` is used (Greilich et al., 2010). The library implements various track structure models, along with range models of the secondary electrons, and provides tools to compute RDDs for different ions and materials. To compute the slowing and stopping of ions in CR-39, stopping power data are generated with `SRIM` with a density of $1.31\,\mathrm{g\,cm^{-3}}$ (Ziegler et al., 2010).

`libamtrack`, however, does not include CR-39 as a predefined material, and the RDD is calculated for water for the given ion and energy. To obtain the RDD for CR-39, two corrections are applied. First, the maximum range of δ-electrons, which sets the outer (penumbra) radius of the RDD, is rescaled by the inverse density ratio $\rho_{\text{water}}/\rho_{\text{CR-39}}$. This scaling is physically motivated: in the continuous slowing down approximation (CSDA), charged-particle ranges scale approximately as $\propto 1/\rho$ for materials of similar effective atomic number, i.e. relevant to liquid water and CR-39. This approximation is widely applied when converting particle ranges between tissue-equivalent materials in particle therapy dosimetry (Janni, 1982). Second, the dose is renormalised by the CR-39/water stopping power ratio obtained from `SRIM` which ensures that the radial integral of the RDD equals the CR-39 LET.

A more rigorous treatment would require either a CR-39-specific parametric RDD model or Monte Carlo simulations of δ-electron transport in CR-39, both of which are left for future work.

Figure 2 illustrates examples of RDDs in CR-39 calculated using the RDD model proposed by Cucinotta et al. (1998) as implemented in `libamtrack` and renormalised to CR-39. Different RDDs will lead to different local dose distributions and thus affect the calculated track formation. The present work focuses on the model by Cucinotta et al. (1998); a study on the effect of different RDD models or numerical RDD simulations across the dataset is left for future work. The comparison between figure 2a–b, showing the RDDs for a single proton and carbon ion, respectively, highlights how the local distribution varies orders of magnitude within a very short range.

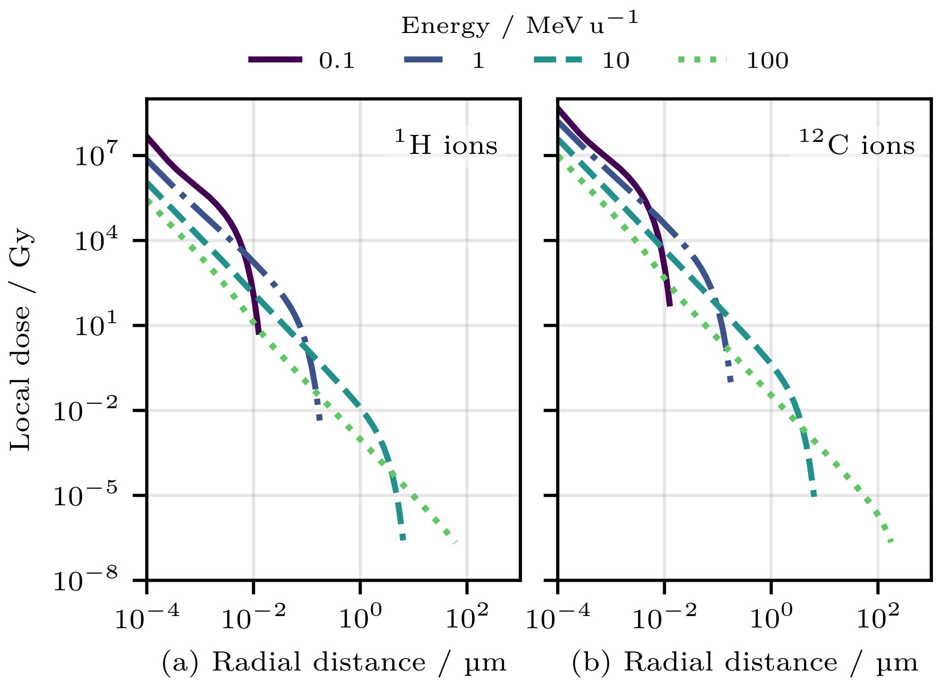


Figure 2: Radial dose distributions in CR-39 for (a) protons and (b) carbon ion tracks using the amorphous track structure model of Cucinotta et al. (1998) plotted for different kinetic energies.

Following the discussion of Katz (1984), it is expected that the δ-electrons will interact with polymer clusters in the CR-39, causing local damage that leads to variations of the local etch-rate.

Dose distribution maps

To model track formation, the dose distribution is computed as the ion slows down in the detector material. This is achieved by calculating the RDD at discrete positions along the ion track while accounting for the gradual energy loss of the ion as it slows down.

Using the CSDA together with stopping-power data from `SRIM`, the ion energy loss is computed at each step, and the corresponding RDD is calculated using the methods described in Section 2.2. By combining the RDDs along the track, a two-dimensional (2D) dose map $d(r, z)$ is constructed, where r denotes the radial distance from the track core and z the position along the track axis, perpendicular to the detector surface.

Figure 3 shows examples of radial dose distributions for different ion species and energies. The top row (a)–(c) illustrates the change of the kinetic energy of the ion along the track axis z as it slows down in CR-39, while the bottom panel (d)–(f) shows the corresponding dose maps. The LET distribution as the ion slows down, plotted with dashed lines, is merely shown for reference and is not used within this work.

Three representative cases are shown: Figure 3(a,d) a 1.0 MeV/u proton, figure 3(b,e) a 1.0 MeV/u carbon ion, and figure 3(c,f) a 10 MeV/u carbon ion, where the latter remains relatively constant throughout the modelled depth of 25 µm of CR-39.

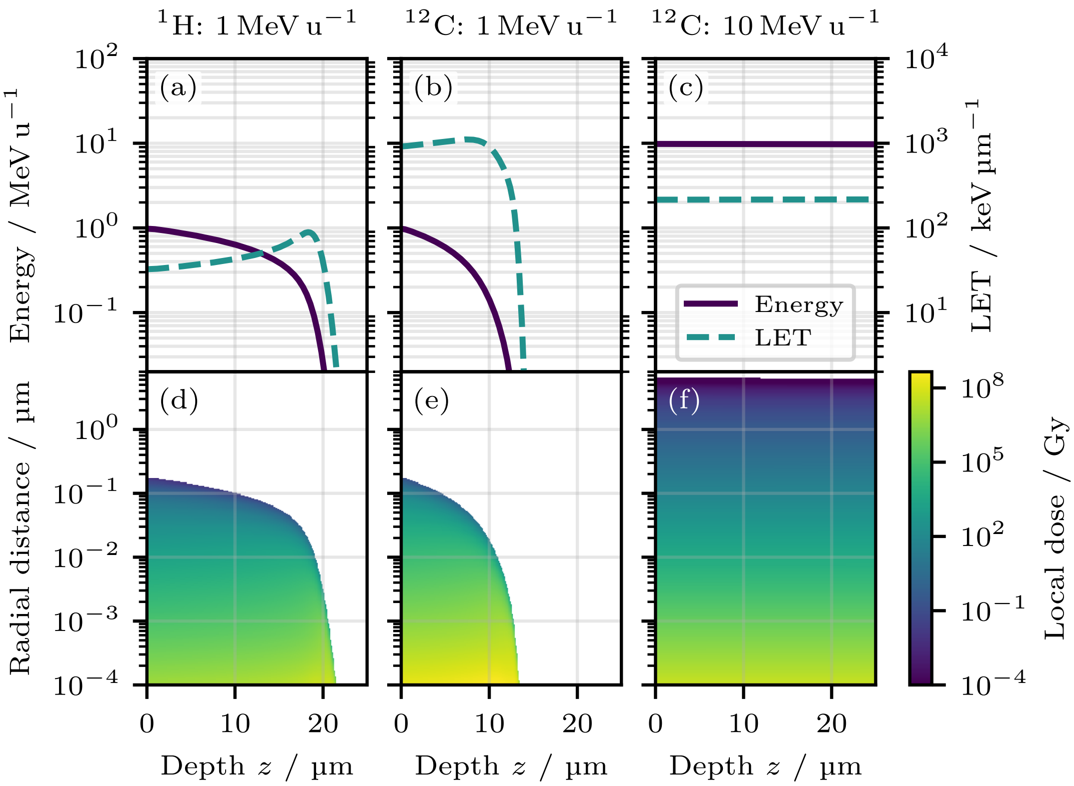


Figure 3: CSDA-calculated energy distribution along the track axis z (solid lines) for a (a) 1.0 MeV/u proton, (b) 1.0 MeV/u carbon ion, and (c) 10 MeV/u carbon ion slowing down in CR-39. The LET along the track is shown by dashed lines for reference. The bottom panel (d)–(f) shows the corresponding radial dose distributions calculated from the kinetic energy along z.

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The dose maps demonstrate the strong variation of the local dose distribution with ion species and energy, which directly influences the resulting etch-rate distribution and track formation.

2.3 Etch-rate prediction

Numerous studies have investigated the relationship between local energy deposition and the resulting etch-rate in CR-39 detectors (Fromm et al., 2019). Although a complete microscopic understanding of how localized absorbed dose translates into polymer damage and etch kinetics is still lacking (Yamauchi, 2003), experimental data on LET-based models consistently shows that the etch-rate increases with increasing energy deposition (Azooz et al., 2012; Dörschel et al., 2003a; Fromm et al., 1991; Nikezic & Yu, 2004).

As a somewhat phenomenological approach, this work assumes that the etch-rate v_d depends directly on the local absorbed dose d in the polymer, i.e. $v_d(d)$. This formulation generalises the conventional track etch-rate description v_t used in LET-based models. The underlying assumption is that polymer damage around the ion trajectory is primarily caused by secondary electrons, which, for a given local dose d, induce similar chemical modifications irrespective of the primary ion species or energy. This also implicitly assumes that effects of differences in the energy spectra of the secondary electrons are neglected, which is a weakness of amorphous track structure theory.

The functional form of $v_d(d)$ is not known and must therefore be determined through calibration against experimental data for a given CR-39 type and etching method. Nevertheless, general features of $v_d(d)$ can be inferred from existing experimental observations of track formation in CR-39 detectors. An idealised illustration of possible $v_d(d)$ behaviours is shown in figure 4.

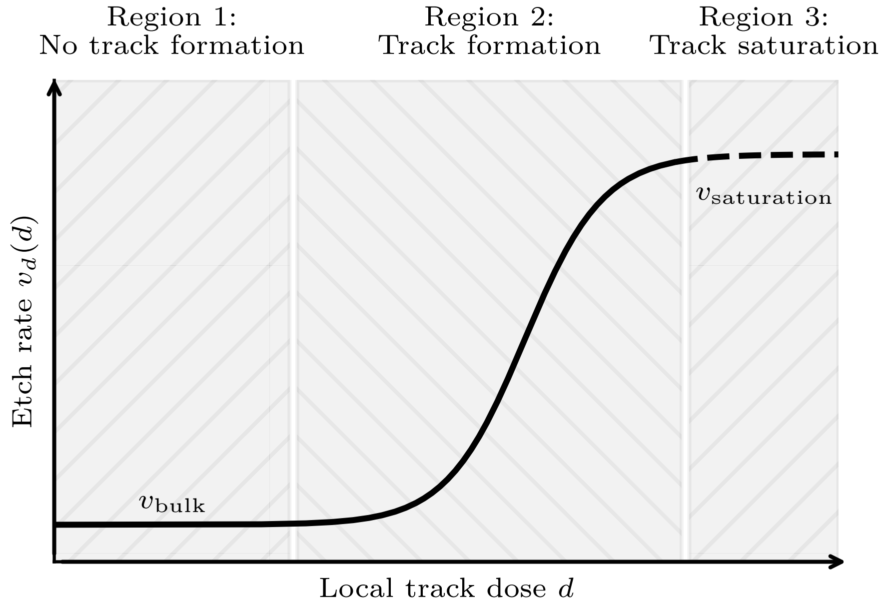

Figure 4: Idealised etch-rate model $v_d(d)$ as a function of the local absorbed dose d in ion tracks. At low doses, the etch-rate converges to the bulk etch-rate v_{bulk}. At very high doses, the etch-rate is expected to saturate at a maximum value $v_{\text{saturation}}$ corresponding to fully damaged polymer material. It is assumed that the same $v_d(d)$ function, for a given CR-39 material and etching method, can be used to describe track formation for all ions.

Based on these considerations, it is assumed that $v_d(d)$ exhibits three distinct regimes:

Region 1 For low local doses, the etch-rate converges to the bulk etch-rate $v_d(d \to 0) = v_{\text{bulk}}$, as the polymer remains largely undamaged. This regime is relevant for sparsely ionising radiation such as photons and fast protons, which do not lead to track formation after etching.

Region 2 For higher-LET particles, including slow protons and heavier ions, the local dose increases substantially, leading to enhanced polymer damage and a corresponding increase in the etch-rate, as widely reported in the literature.

Region 3 For extremely high local doses, the etch-rate is expected to saturate at a maximum value $v_{\text{saturation}}$, reflecting the finite extent of polymer damage. This regime may become relevant for ions heavier than carbon, e.g. fission fragments.

These assumptions are based on several studies that have investigated the response of CR-39 to gamma irradiation at doses exceeding the kGy range, showing a relationship between the etch-rate v and gamma dose (Singh & Neerja, 2007). Using ethanol potassium hydroxide (KOH), Brahimi et al. (2008) mapped the etch-rate as a function of gamma dose and observed an approximately constant etch-rate at doses up to kGy-levels, followed by an increase at increasing dose. Similarly, Yamauchi et al. (2021) reports a dose threshold of approximately 10 kGy, above which gamma irradiation significantly affects the etch-rate properties. This suggests a transition from Region 1 to Region 2 above kGy-level doses.

As visualized in figure 2a, the RDDs provide insight into the plausible shape of $v_d(d)$. For example, the local dose produced by a 100 MeV u^{-1} proton, which generally does not generate a latent track, is roughly two orders of magnitude lower than that of a 1.0 MeV u^{-1} proton, which does form a latent track. Indeed, the local dose of the 100 MeV u^{-1} proton drops below kGy-levels outside a 1 nm radius, often considered to be the dimension of the track core region.

In terms of classical detector-response modeling, the detector response (i.e. etch-rate) can be fully characterized experimentally as a function of gamma dose and subsequently folded with the RDD. This approach is in line with the original methodology proposed by Katz (1984). However, for slow, heavier ions (see figure 2b), the local doses exceed hundreds of MGy, beyond experimentally accessible ranges. Therefore, the shape of $v_d(d)$ in Regions 2 and 3 must be inferred from track formation data for different ion species and energies. A prerequisite for this model is that the bulk etch-rate v_{bulk} under the given etching conditions is known and constant.

Debris-damped etching in the track core

The dose-based model $v_d(d)$ captures how local polymer damage accelerates etching, but it does not explicitly account for transport limitations inside narrow and deep etch pits (Nikezic & Yu, 2004). In practice, reaction products and detached polymer fragments may accumulate in the pit which limits the amount of fresh etchant diffusing into it. This can reduce the effective etch-rate near the track core, in particular at large pit aspect ratios.

To account for this effect phenomenologically, a damping factor η is introduced and the effective etch-rate map is written as

$$v_{\text{eff}}(r, z) = v(r, z)\, \eta(r, z), \tag{1}$$

with $0 < \eta \leq 1$. The damping factor is parameterised as

$$\eta(r, z) = \frac{1}{1 + \left(\frac{A(r,z)}{\alpha}\right)^{\beta}}, \tag{2}$$

where $A(r, z) = z/r_{\text{track}}(z)$ is the local aspect ratio of the etch pit, defined as the ratio of depth z to the etch-pit radius $r_{\text{track}}(z)$ at that depth. Thus, parameter α defines the characteristic aspect-ratio scale at which damping becomes significant, while β controls the transition steepness. The effective etch-rate v_{eff} is bounded below by v_{bulk}, ensuring that damping cannot reduce the etch-rate below the undamaged bulk value.

In this formulation, $v_d(d)$ and the debris-damping parameters α and β are calibrated simultaneously against track-shape data below. The effective speed map $v_{\text{eff}}(r, z)$ is then used in the subsequent eikonal-based contour calculations.

2.4 Track contour calculation

To model track formation, the spatial etch-rate distribution $v(r, z)$ is interpreted as a speed map for the advancing etching front, analogous to a wave front propagating through a medium with spatially varying speed. It is assumed that the etching front begins at the detector surface at time $t = 0$ and propagates into the detector with a local speed given by $v_{\text{eff}}(r, z)$. The arrival time of the etching front at each point, $t(r, z)$, can be computed by solving the eikonal equation (Sethian, 1999)

$$|\nabla t(r, z)| = \frac{1}{v_{\text{eff}}(r, z)}, \tag{3}$$

where $t(r, z)$ denotes the arrival time at position (r, z). Iso-contours of $t(r, z)$ define the track contour at a given etching time. This formulation allows track contours to be computed for arbitrary etching times and spatially varying etch-rate distributions.

Several algorithms exist to solve this equation. On structured grids, the Fast Marching Method can be used (Sethian, 1999). However, to accurately resolve the steep dose gradients near the track core, an adaptive unstructured grid is required. In this work, Dijkstra's algorithm (Dijkstra, 1959) is applied on such a grid, providing fine spatial resolution in regions of high local dose near the track core.

Figure 5 illustrates an example of track contour calculation for a $1\,\text{MeV}\,\text{u}^{-1}$ carbon ion using a functional form of $v_d(d)$ not calibrated to experimental data. Figure 5a shows the etch-rate map $v(r, z)$, representing the local speed of the etching front, with the fastest paths and travel paths to two points a few microns apart highlighted. Figure 5b shows the corresponding arrival times $t(r, z)$ obtained by solving the eikonal eq. (3) discretely with Dijkstra's algorithm. To illustrate track contours, iso-arrival time contours for etching times of $t = 2\,\text{h}$ and $t = 5\,\text{h}$ are shown, with the corresponding track radii calculated and annotated.

In this example, the ions impinge perpendicular to the detector surface, resulting in cylindrically symmetric track contours. The method can be extended to arbitrary angles of incidence by initializing the etching front with a corresponding rotation, out of the scope of this work.

2.5 Experimental data for validation

To calibrate the etch-rate $v_d(d)$ and benchmark the proposed track formation model, the extensive experimental data available in Dörschel et al. (2003a) are used. The study reports measurements of track shapes in CR-39 detectors for four ions as listed in table 1. The track shapes are measured for two different etching times, providing eight unique datasets. Dörschel et al. (2003a) provides a bulk etch-rate $v_{\text{bulk}} \equiv v_d(d = 0) = 1.73\,\mu\text{m}\,\text{h}^{-1}$.

Through the definition of $v_d(d)$, a single etch-rate function can be calibrated to reproduce the track shapes for all particle-energy combinations.

Outside the scope of this manuscript, Dörschel et al. (2003b) provides similar measurements of track shapes for oblique tracks to be used for future model validation at non-perpendicular incidence.

3 Results and Discussion

The model framework outlined in figure 1 and detailed in the previous sections is implemented in `python`. The objective is to find the unique etch-rate function $v_d(d)$ that

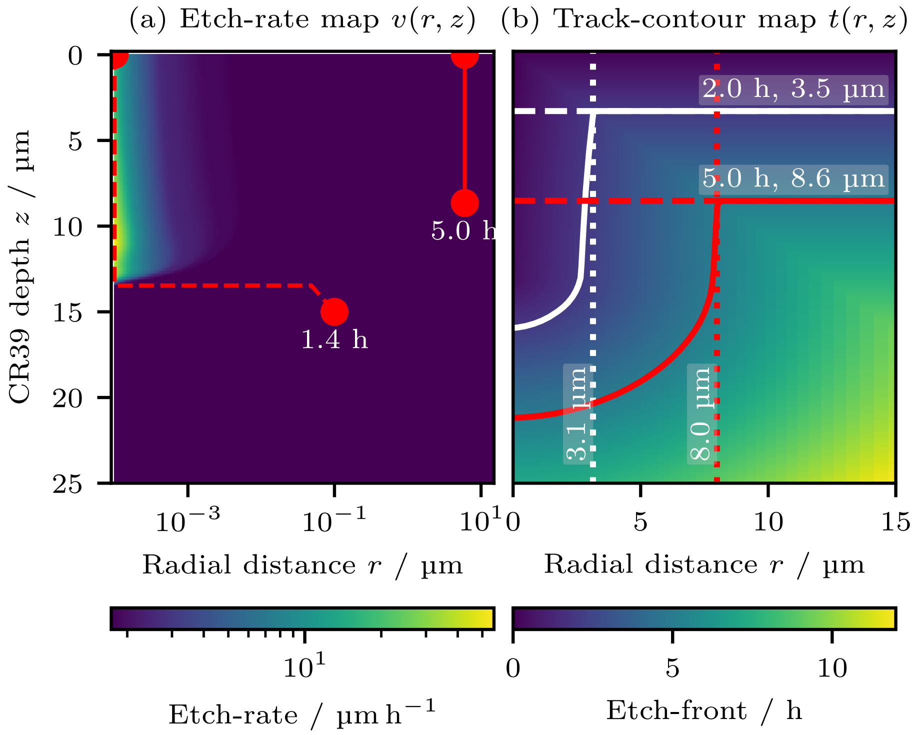

Figure 5: Example of track contour calculation for a 1 MeV u^{-1} carbon ion (CSDA range 13 µm) using an arbitrary $v_d(d)$. (a) Etch-rate map $v(r, z)$ representing the local speed of the etching front, including the fastest paths to two spatial points to illustrate the propagation of the etching front and the time it takes to reach them. (b) Arrival times $t(r, z)$ obtained by solving eq. (3) for the speed map in (a). Iso-arrival time contours for $t = 2$ h (solid, white line) and $t = 5$ h (solid, red line) are highlighted. Both calculated radii are and The etch-rate map is plotted on a logarithmic scale to better visualize the high speeds near the track core.

Table 1: Particle and energy combinations used for model calibration provided by Dörschel et al. (2003a). The first three columns are given in Dörschel et al. (2003a), the CSDA range and LET in CR-39 are calculated for reference.

Particle	Etch times / h	Energy / MeV	Energy / MeV u^{-1}	CSDA$_{\text{CR-39}}$ / µm	LET$_{\text{CR-39}}$ / keV µm^{-1}
^{1}H	6.0, 12	1.27	1.26	28.6	28.2
^{4}He	3.0, 6.0	4.00	0.999	20.9	134
^{7}Li	0.83, 1.50	4.82	0.694	12.3	362
^{12}C	0.61, 1.00	14.8	1.23	16.2	832

can reproduce the experimental track shapes for all particle-energy combinations listed in table 1. Specifically, the framework takes as input the ion species, kinetic energy, and etching time, and outputs the predicted track contour.

3.1 Etch-rate calibration

It is assumed that the shape of $v_d(d)$ depends on the etching solution and etching temperature, but is independent of particle type and energy. Accordingly, $v_d(d)$ must be calibrated for the specific etching procedure used in the experiments. For this purpose, $v_d(d)$ is calibrated to match the etching procedure reported by Dörschel et al. (2003a) with the bulk etch-rate $v_d(0) = 1.73\,\mu\mathrm{m\,h^{-1}}$.

Figure 6a shows the frequency distribution $f(d)$ of local doses in the tracks for the particles listed in table 1, illustrating the range of local doses produced by different ion species and energies. The available experimental data does not provide sufficient information to constrain the shape of $v_d(d)$ at very high local doses above 100 MGy, where saturation of the etch-rate could occur. Constraining this high-dose regime would require track shape data for much heavier ions, such as iron or xenon, which is left for future work.

Based on the considerations discussed in section 2.3, a piecewise-linear function in log-log space is chosen as the initial parametrisation for the optimisation, providing flexibility to capture the expected behaviour in the three dose regimes illustrated in figure 4. The function is defined by dose-etch-rate anchor points, and the anchor velocities are calibrated against all track-shape datasets in (Dörschel et al., 2003a) simultaneously using a least-squares optimisation procedure.

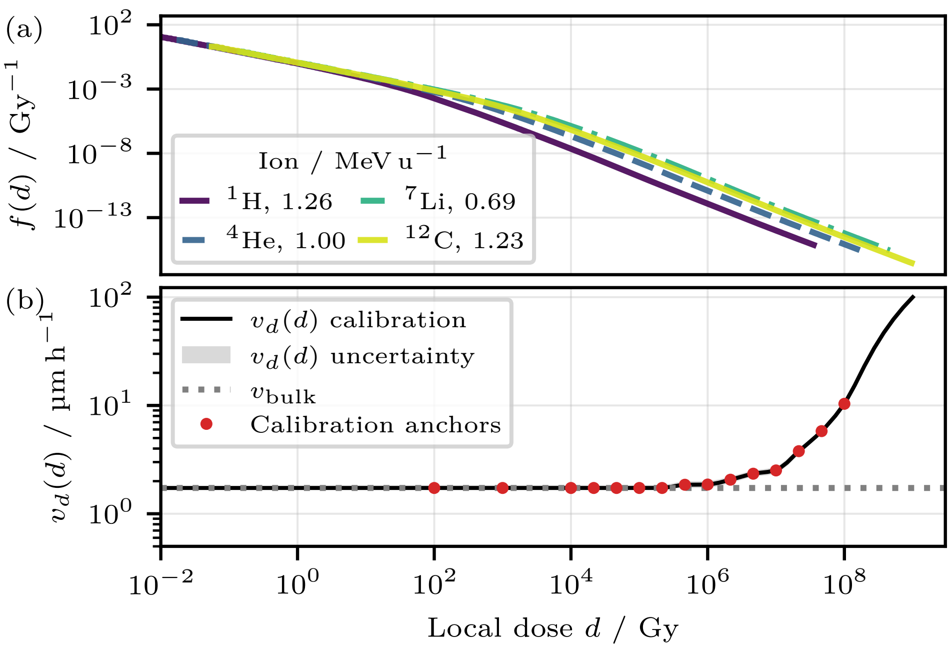

Figure 6: (a) Calculated frequency distribution $f(d)$ of local doses in the ion tracks for the particles listed in table 1. (b) Etch-rate $v_d(d)$ as a function of local dose, optimised to reproduce the experimental track shapes reported in table 1.

Figure 6b shows the resulting etch-rate function $v_d(d)$ as a function of local dose, optimised to reproduce the experimentally measured tracks reported in table 1. The uncertainty band is derived from the fit uncertainties of the model parameters. The resulting $v_d(d)$ exhibits the expected behaviour, with a low-dose regime in which $v_d(d)$ remains close to the bulk etch-rate up to approximately 1 MGy, followed by a steep increase at higher doses.

3.2 Track formation calculations

The unique etch-rate relationship $v_d(d)$ identified in the previous section is used to calculate track formation for the particle-energy combinations listed in table 1. The results are shown in figure 7, where the predicted track contours (lines) are compared with the experimental data (markers) for the two etching times associated with each particle-energy combination. The shaded regions in figure 7 indicate the resulting uncertainty in the predicted track contours derived from the fit uncertainties of the etch-rate model parameters.

Quantitative comparisons of predicted versus measured track radii and depths for the validation dataset are summarised in table 2, including uncertainties for both experimental and simulated values. This demonstrates that a single etch-rate function $v_d(d)$ (figure 6b), obtained through minimisation of the experimental data and the corresponding simulated track formation data (figure 7), can reproduce the track shapes for all particle-energy combinations in agreement within the uncertainties.

Table 2: Predicted and measured track radii and depths for the model validation dataset. Measured values are from Dörschel et al. (2003a) for the specified etching times; the experimental uncertainties of 0.30 µm on track radii and 0.60 µm on track depths are as stated therein. Predicted values are computed using `tracketch` with the optimised $v_d(d)$ shown in figure 6b and are likewise reported with their estimated ($k = 1$ coverage factor) uncertainties.

Particle	Energy / MeV u^{-1}	Etch time / h	Track radius / µm		Track depth / µm	
			Measured	Simulated	Measured	Simulated
^{1}H	1.26	6.0	3.3 ± 0.30	3.0 ± 1.2	14 ± 0.60	14 ± 3.2
^{1}H	1.26	12	8.3 ± 0.30	6.8 ± 2.2	31 ± 0.60	33 ± 0.94
^{4}He	1.0	3.0	2.8 ± 0.30	3.0 ± 0.10	13 ± 0.60	15 ± 1.2
^{4}He	1.0	6.0	6.8 ± 0.30	6.3 ± 0.17	25 ± 0.60	26 ± 0.27
^{7}Li	0.69	0.83	1.1 ± 0.30	1.1 ± 0.020	8.8 ± 0.60	8.5 ± 0.51
^{7}Li	0.69	1.5	1.8 ± 0.30	2.0 ± 0.020	13 ± 0.60	13 ± 0.050
^{12}C	1.23	0.61	0.76 ± 0.30	0.92 ± 0.010	11 ± 0.60	13 ± 0.69
^{12}C	1.23	1.0	1.6 ± 0.30	1.5 ± 0.010	19 ± 0.60	17 ± 0.62

3.3 Track geometry predictions

With the etch-rate function $v_d(d)$ calibrated to the experimental data, the model can be used to predict track geometries for arbitrary particle types and energies. Figure 8 shows the predicted track minor axis radii as a function of ion energy for the ions used for calibration, as well as for ^{9}Be and ^{11}B, which were not included in the calibration dataset. The selected etching times include those reported in table 1, enabling direct

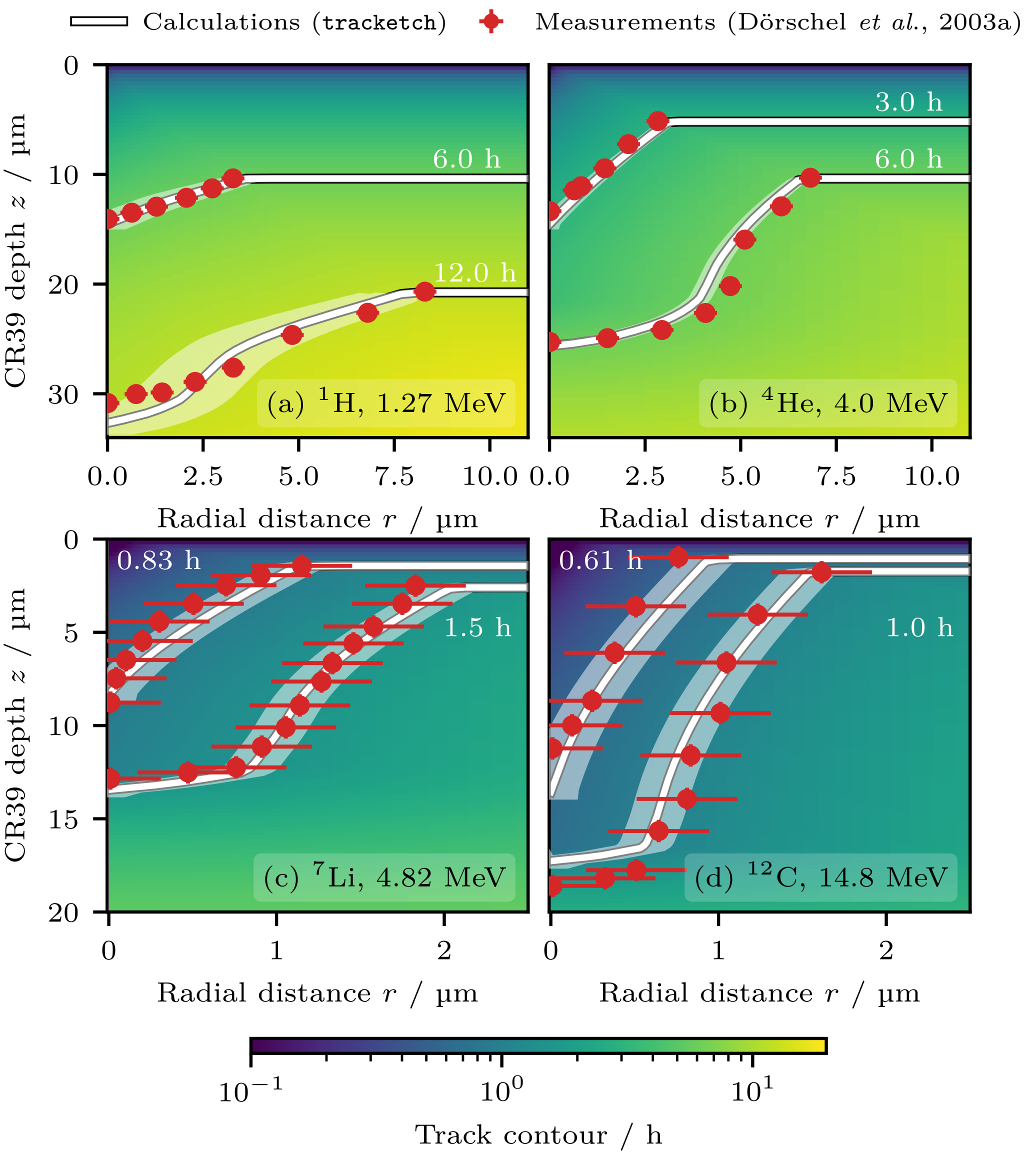

Figure 7: Validation of calculated track shapes for the particle-energy combinations listed in table 1: (a) 1.27 MeV proton, (b) 4.0 MeV helium ion, (c) 4.82 MeV lithium ion, and (d) 14.8 MeV carbon ion. For each case, the calculated arrival-time contours, i.e. representing the predicted track shapes, are compared with experimental data for two etching times. The solid lines indicate the track contours obtained using the optimised $v_d(d)$ function shown in figure 6b. The reported experimental uncertainties in Dörschel et al. (2003a) are not large enough to be visible in all subfigures. The shaded bands indicate the uncertainty in the predicted track contours.

comparison in the calibrated regime while also showing predictions for ions not used in the fit. For reference, the calculations for protons also include results from `TRACK_P`, where the two etch-rate model is calibrated for protons only (Nikezic et al., 2016).

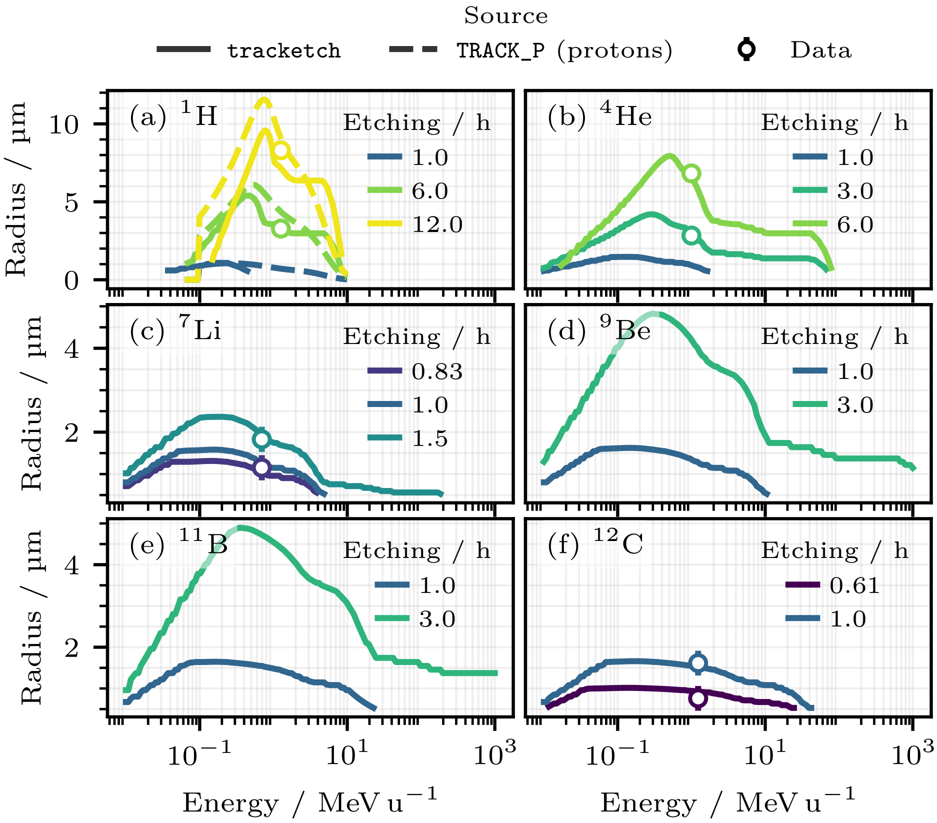

Figure 8: Predicted track minor axis radii as a function of ion energy for various particle species and etching times. Panels (a)–(f) show the effect of different etching time and kinetic energy for ions from protons up to carbon, including the calibration-relevant times listed in table 1. The measurements from Dörschel et al. (2003a) are provided as markers for reference where available. Besides the ions used for calibration, ^{9}Be and ^{11}B are included to illustrate predictive capability beyond the fitted dataset. For protons, the track radii are calculated using `TRACK_P` for reference (Nikezic et al., 2016). All predictions are based on the optimised etch-rate function $v_d(d)$ shown in figure 6.

As oblique tracks will appear as ellipses, which per definition mainly increases the major axis length, the minor axis diameter is used as a more robust measure of track size for future validation against oblique track data. The predicted track radii exhibit the expected behaviour, with larger track sizes for heavier ions and lower energies, as well as for longer etching times. It appears that ions with a kinetic energy of approximately 0.5 MeV u^{-1} produce the largest track radii, which approximately amounts to the energies at the Bragg peak region.

3.4 Limitations and future work

While the proposed track formation model provides a novel algorithm to predict track shapes for various ion species and energies, several limitations and tasks for future work are identified:

Amorphous track structure approximation: The adopted amorphous track structure framework is a coarse-grained description of energy deposition, developed for other materials than CR-39, and does not resolve ultra-fine, material-specific modifications in CR-39 (e.g., nanometre-scale chemical and structural heterogeneity along the latent track). This could be improved using numerical track structure simulations that explicitly model the transport of secondary electrons and their interactions with the polymer.

Model assumptions: The model relies on the assumption that the etch-rate depends solely on the local absorbed dose, neglecting potential dependencies on the energy spectra of secondary electrons or other factors. Future work could explore more complex dependencies of the etch-rate on additional parameters.

High-dose regime: The shape of $v_d(d)$ at very high local doses remains unconstrained due to the lack of experimental data for heavier ions. Future experiments with heavy ions could provide data to constrain this regime and validate the predicted saturation behaviour.

Oblique incidence: The current model is limited to ions impinging perpendicularly onto the detector surface. Rotating the plane for which the eikonal equation is solved would allow the model to be extended to arbitrary angles of incidence, which is relevant for many applications. Several sources, e.g. Dörschel et al. (2003b), provide experimental data for oblique tracks that can be used for validation of the model at non-perpendicular incidence.

Missing track effects: Yamauchi et al. (1995) report the phenomenon of missing tracks for low-energy deuterons in (TS-16) CR-39 detectors for an etching solution of 6 M KOH solution at 70 °C, which differs from the one used in this work. This phenomenon could be investigated with `tracketch` by re-calibrating $v_d(d)$ for the specific etching conditions and comparing the predicted track shapes with experimental data for deuterons.

Etching condition dependence: The model is calibrated for a specific etching solution and temperature for a constant v_{bulk}. Variations in etching temperature, etchant concentration, buffer composition, and detector batch properties are expected to affect both the bulk etch-rate and the shape of $v_d(d)$, and remain to be systematically characterized.

Conclusion

This work presents a novel open-source framework, `tracketch`, to predict track formation for ions impinging on CR-39 detectors. In contrast to previous track formation models that model the etch-rate v_t as a function of residual range or different types of restricted linear energy transfer, the proposed model directly links the etch-rate to the local absorbed dose d, i.e. $v_d(d)$.

Amorphous track structure theory is used to calculate the local dose distribution around ion trajectories, and a single dose-dependent etch-rate function $v_d(d)$ is identified to model how local polymer damage translates into etching kinetics. Modelling the etch-rate as a function of local dose provides a more physically motivated description of track formation, and allows for prediction of track formation outside the ions used for calibration, as the local dose distribution can be calculated for arbitrary ion species and energies.

To calculate the track shape after a certain etching time, the advancing etching front is interpreted as a wave front propagating through a medium with spatially varying speed. The any point, the local speed is given by the etch-rate $v(r, z)$ calculated from $v_d(d)$. The resulting iso-arrival-time contours correspond to the predicted track geometry for a given ion species, energy, and etching time.

`tracketch` is calibrated against experimental data for various ion species and energies by benchmarking the predicted track geometries against measured track shapes. Using a single etch-rate function $v_d(d)$ for all ion species, the model reproduces the track shapes for all particle-energy combinations in agreement with the experimental data within the uncertainties. It is demonstrated how the model can be used to predict track morphology of ions not used in the calibration.

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Code example

Any users of `tracketch` is referred to the documentation for installation instructions and usage examples[2]. A brief example of how to use `tracketch` to calculate arrival-time maps and track contours is shown below, which more examples available online. The calculation of track contours for a given ion species, energy, and etching time takes 0.1 s on a standard laptop, depending on the desired spatial resolution and particle range.

Listing 3.1: Example usage of `TrackSimulator` forplotting and track shape calculation.

```python
from tracketch import TrackSimulator

# create a simulator for 100 MeV/u carbon ions
sim = TrackSimulator(
    particle_name="12C",
    start_energy_MeV_u=100.0,
)

# calculate etch front arrival-time map
fig, ax = sim.plot_map(name="arrival")

```

[2]Available at https://github.com/jbrage/tracketch

```
# plot and get the track contour for 1 hour of etching
r_um, z_um = sim.plot_iso_time_contour(ax=ax, etching_time_h=1.0)
```

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